

Orientation – Day 4

BATCH 22

Types of Software

- 1. System Software:** This type of software includes operating systems, device drivers, utilities, and other programs that enable the hardware to function and provide a platform for running application software. Examples include Windows, macOS, Linux, iOS, and Android.
- 2. Application Software:** Application software is designed to perform specific tasks or solve particular problems for end-users. It includes a wide range of programs such as word processors, spreadsheets, web browsers, email clients, multimedia players, and games.

3. Programming Software: Programming software provides tools for developers to write, debug, and maintain software applications. Integrated Development Environments (IDEs), text editors, compilers, interpreters, and debuggers are examples of programming software.

4. Utility Software: Utility software consists of programs designed to help manage and maintain the computer system efficiently. This includes antivirus software, disk cleaners, backup software, file management tools, and system optimization utilities.

Integrated Development Environment

- IDE stands for Integrated Development Environment.
- It's a software application that provides comprehensive facilities to programmers for software development.
- An IDE typically includes a source code editor, build automation tools, debugging tools, and other features that help streamline the process of writing, testing, and debugging software.
- IDEs are often tailored to specific programming languages or platforms, providing developers with a cohesive environment for writing code and managing projects.

BEST IDE FOR PROGRAMMING

- Jupiter Notebook
- Android Studio
- Code block
- NetBeans
- Pycharm
- Eclipse
- Visual Studio

Jupyter Notebook

- Jupyter Notebook is a web-based interactive computing platform.
- However, Jupyter Notebook integrates well with several popular IDEs, and there are ways to enhance its functionality to provide a more IDE-like experience.
- Open-source web application for creating and sharing interactive documents that combine:
 - Live code (Python, R, Julia, etc.)
 - Rich text
 - Equations
 - Visualizations
 - Multimedia elements

- Focuses on
 - Exploratory data analysis
 - Data visualization
 - Rapid prototyping
- Not designed for large-scale software development projects.

- While Jupyter Notebook itself isn't a full-fledged IDE, it can be integrated with several popular IDEs to add their features
 1. **PyCharm:** Powerful Python IDE with good Jupyter Notebook integration for code completion, debugging, and variable exploration.
 2. **Visual Studio Code:** Versatile code editor with various extensions to enhance Jupyter Notebook functionality, like variable management and code navigation.

Pycharm

- PyCharm is a powerful IDE (Integrated Development Environment) specifically designed for Python development by JetBrains. It's a popular choice among Python programmers due to its extensive features and functionalities that streamline the development process
- 1. Intelligent Code Assistance**
 - 2. Advanced Debugging Tools**
 - 3. Built-in Unit Tester**
 - 4. Version Control Integration**
 - 5. Cross-Platform Compatibility**

Free vs. Professional Edition

- PyCharm offers two main editions:
- **Community Edition (Free):** Open-source and free to use, providing core features essential for Python development.
- **Professional Edition (Paid):** Offers additional features like advanced code analysis, web development frameworks debugging, database tools, scientific computing support with Jupyter Notebook integration, and remote development capabilities.

Visual Studio

Visual Studio (VS) is a powerful and versatile Integrated Development Environment (IDE) from Microsoft. It's a popular choice for a wide range of software development projects due to its comprehensive features and support for various programming languages

- C#
- C++
- F#
- Visual Basic
- Python
- JavaScript

- **Web Development Support:**

- Supports development for web applications and services using frameworks like ASP.NET, ASP.NET MVC, and Node.js.
- Provides built-in tools for web development tasks like HTML editing, CSS styling, and JavaScript development.

- **Mobile Development:**

- Enables development for various mobile platforms, including Android, iOS, and Windows mobile (limited support).
- Integrates with tools like Xamarin for cross-platform mobile app development.

Free vs. Paid Editions

Visual Studio offers several editions with varying feature sets:

- **Visual Studio Community (Free):** A fully-featured free edition suitable for individual developers, open-source projects, and learning purposes. It offers most core functionalities for various development needs.
- **Visual Studio Professional (Paid):** Provides additional features for professional development, like increased debugging capabilities, code performance analysis tools, and more advanced web development support.
- **Visual Studio Enterprise (Paid):** The most comprehensive edition, catering to large-scale enterprise development with features like architecture tools, performance optimization tools, and enhanced collaboration functionalities.

XAMPP Server – Local Web Hosting

- The most basic tool that you need to set up your web development environment
- XAMPP is a free open source cross-platform web server solution stack package, consisting mainly of the Apache HTTP server, My SQL database, and interpreters for scripts written in the PHP and Perl Programming Languages

Etymology

- XAMPP's name is an acronym for:
- X (to be read as "cross", meaning cross-platform)
- Apache HTTP Server
- MySQL
- PHP
- Perl
- Tomcat

USES

- XAMPP's designers intended it for use only as a development tool, to allow website designers and programmers to test their work on their own computers without any access to the Internet.
- To make this as easy as possible, many important security features are disabled by default.
- XAMPP is sometimes used to actually serve web pages on the World Wide Web.
- A special tool is provided to password-protect the most important parts of the package.
- LAMPP is one of the part in XAMPP.

Apache HTTP Server

- Apache HTTP Server, often referred to simply as Apache, is a free and open-source web server software. It's one of the most popular web servers globally, powering a significant portion of the websites you visit on the internet

Function

- Apache acts as an intermediary between web browsers and web servers.
- When you enter a website address (URL) in your browser, it sends a request to the web server hosting that website.
- Apache receives the request, retrieves the relevant web pages and files from the server, and transmits them back to your browser in a format your browser can understand and display.

Common Use Cases

- **Web Hosting:** Apache is widely used by web hosting providers to host websites for their customers.
- **Personal Websites:** Individuals and small businesses can use Apache to host their own websites on their personal servers.
- **Intranet Applications:** Apache can also be used to serve internal web applications within a company network.

MySQL

- It is the world's most popular open source database. It is a Relational Database Management System (RDBMS) - data and its relationships are stored in the form of tables that can be accessed by the use of MySQL queries in almost any format that the user wants.

PHP

- **PHP** is a server-side scripting language designed for web development but also used as a general-purpose programming language.
- Originally created by Rasmus Lerdorf in 1995, the reference implementation of PHP is now produced by The PHP Group. While PHP originally stood for *Personal Home Page*,[[]it now stands for *PHP: Hypertext Preprocessor*.

PHP

- A server-side programming language which is used to produce dynamic web pages. PHP code can be embedded within HTML. It is also free and platform-independent which means that it can be installed on any operating system.

Client – Server Architecture

- Client-server architecture is a fundamental design pattern used in network computing. It separates the tasks and workloads between two distinct types of programs:
- **Clients:** These are programs that initiate requests for resources or services. They typically run on user devices like personal computers, laptops, mobile phones, or web browsers.
- **Servers:** These are programs that provide resources or services to clients upon request. They are typically more powerful machines with significant storage capacity and processing power. Servers are often located on a central network, although they can also reside on the same system as the client in some cases.

Steps

1. Client initiates request:

The client program initiates a request for a specific resource or service. This request could be anything from fetching a web page to accessing data stored on a database server.

2. Request travels over network:

The request travels across a network, like the internet, using a communication protocol like HTTP (Hypertext Transfer Protocol) or another protocol designed for the specific service.

3. Server processes request:

The server receives the request and processes it. This might involve retrieving data from storage, performing calculations, or interacting with other systems.

4. Server sends response:

The server sends a response back to the client. This response could be the requested data, the results of an operation, or an error message.

5. Client receives and displays response:

The client program receives the response from the server and interprets or displays it for the user.

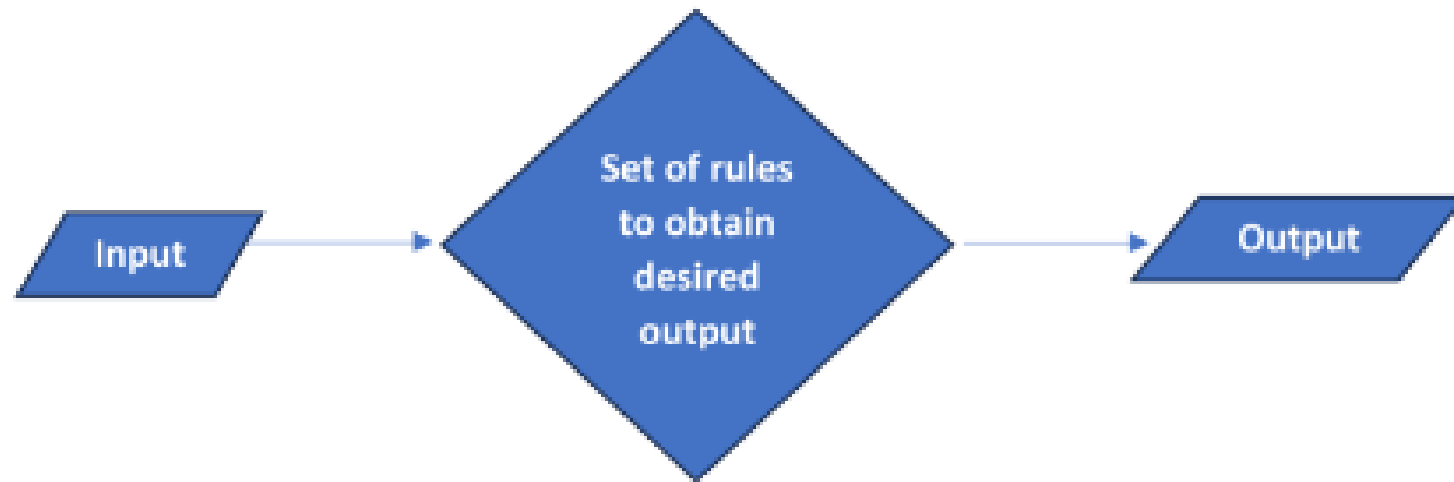
Examples of Client-Server Architecture

- **Web browsing:** When you visit a website in your web browser, it acts as the client, and the web server that hosts the website acts as the server.
- **Email:** When you send or receive email, your email client software acts as the client, and the email server acts as the server.
- **Online banking:** When you access your online bank account, your web browser acts as the client, and the bank's secure server acts as the server.

Algorithm

- An algorithm, in the context of computer science and mathematics, is a set of well-defined instructions that provide a step-by-step process for solving a problem or performing a specific task.
- It's essentially a recipe for a computer to follow, breaking down complex problems into smaller, manageable steps

Algorithm



Algorithm

- **Algorithm** is not the complete code or program, it is just the core logic(solution) of a problem, which can be expressed either as an informal high level description as pseudocode or using a flowchart.

Algorithm

Every Algorithm must satisfy the following properties:

- **Input-** There should be 0 or more inputs supplied externally to the algorithm.
- **Output-** There should be atleast 1 output obtained.
- **Definiteness-** Every step of the algorithm should be clear and well defined.
- **Finiteness-** The algorithm should have finite number of steps.

Flowchart

- A flowchart is a visual representation of a process, workflow, or algorithm. It uses a standardized set of symbols and connecting arrows to depict the steps involved and the sequence in which they occur.

Symbol	Name	Function
	Start/end	An oval represents a start or end point
	Arrows	A line is a connector that shows relationships between the representative shapes
	Input/Output	A parallelogram represents input or output
	Process	A rectagle represents a process
	Decision	A diamond indicates a decision

Pseudocode

- Pseudocode is a way of describing the steps in an algorithm in a format that's easy for humans to understand. It uses a mix of keywords and natural language that resembles a programming language, but it's not intended to be compiled or run by a computer directly.

Common Elements in Pseudocode

- **Keywords:** Basic keywords like IF, THEN, ELSE, WHILE, FOR, etc. are often used to represent control flow statements.
- **Variables:** Variables are used to store data and can be named descriptively to reflect their purpose.
- **Functions:** Complex operations can be encapsulated in functions for better organization and reusability.
- **Comments:** Comments can be included to explain specific parts of the algorithm or provide additional clarity.

Example of pseudocode calculating the factorial of any number

```
function factorial(number)
  /* This function calculates the factorial of a number */
  IF number < 0 THEN
    RETURN "Error: Factorial is not defined for negative numbers"
  ELSE IF number == 0 THEN
    RETURN 1
  ELSE
    fact := 1
    FOR i := 1 TO number
      fact := fact * i
    RETURN fact
  END IF
END function
```

Command-line interface (CLI),

- The command-line interface (CLI), also known as command prompt or terminal, is a text-based interface for interacting with your computer's operating system.
- Here are some basic commands commonly used in various CLI environments

Navigation

1. `cd` (change directory): This command allows you to navigate between directories in your file system.

- Example: `cd Documents` - Changes the current directory to "Documents".
- Example: `cd ..` - Moves you up one level in the directory structure.

2. `pwd` (print working directory): This command displays the full path of the current working directory you're in.

3. `ls` (list): This command lists the contents of the current directory, including files and subdirectories.

- You can use options like `-l` for long listing format with detailed information or `-a` to show hidden files.